Ram Calculus Audit of the Catalan Irrationality Preprint

Signed Mixed Moments, Contact Identities,
and Exact Barrier Certificates

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Abstract

We study the actual nine-file source of the pinned preprint *Catalan's constant is irrational*. The source theorem is distinguished from exact local moment identities, imported classical tools, and conditional arithmetic-to-analytic implications. Signed mixed moments and Taylor contact are reconstructed on their actual polynomial spaces; neither positivity nor arithmetic nonvanishing is inferred from a coordinate representation. The real-place reduction and its uniform energy regularization are explained with endpoint and diagonal terms retained. Exact rational barrier calculations replay all 43 stationary-point brackets, 50 point bounds, and both norm bounds. An independent Bernstein sign certificate exhausts the stationary points. A curvature estimate reduces each bracket allowance to $2 \cdot 10^{-11}$, giving a same-trial energy upper bound -2.29110250727, versus the source's -2.290939875. Private finite-moment and Bernstein passages are used locally, with explicit domain restrictions and no global irrationality endorsement. Contribution assessment: C2/5, source-specific finite certificates and bounded analytic refinements without certification of the global irrationality claim.

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1 Exact Target and Claim Discipline

The source [1] asserts that

$$G = \beta(2) = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)^2} = L(2, \chi_{-4}) \notin \mathbb{Q}.$$

This report does not adopt that assertion as an established theorem. The authoritative local input is the exported public repository snapshot `openai/math` at commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The manuscript directory is `preprints/Catalans-constant-is-irrational-September-24-2026`. All nine actual TeX files were read: `build/main.tex` and the eight files in `build/sections`: `introduction.tex`, `foundations.tex`, `odd-primes.tex`, `nonvanishing.tex`, `real-place.tex`, `energy.tex`, `certificate.tex`, and `conclusion.tex`. Bibliographic and finite-data evidence supplement these reads; summaries are not substitutes for mathematical source.

Status	Meaning
Source claim	Irrationality, arithmetic nonvanishing, valuation bounds, and geometric consequences as asserted in the pinned manuscript.
Local result	An identity or finite certificate proved or replayed here, with its quantified domain and explicit evidence.
Classical method	Taylor's theorem, Chebyshev identities, determinant integration, Hardy-space compression, Bernstein enclosures, Descartes' rule, and rational logarithm bounds; no novelty is claimed for these.
Conditional bridge	Promotion from bounded calculations to the source's asymptotic determinant contradiction only when its remaining hypotheses have separately been established.

Applicability L4 records a strong match to the finite moment and barrier steps. It is not a contribution score and does not certify the source theorem. C2 describes the bounded contribution of this report.

2 Actual Private-Corpus Interfaces

In the revised TR250, the passage "What a Ram representation means in this report" defines $m = Ec$, $E_{ij} = \ell_i(\phi_j)$ on a specified finite physical space. A square inverse requires invertibility; a rectangular bank requires full column rank and a left inverse on its range. The "concrete polynomial Ram construction" proves a positive Gram matrix for a positive almost-everywhere weight on an interval, not for an arbitrary signed mixed matrix. This distinction is essential here.

Volume 16 supplies the power-to-Bernstein identity, convex-hull enclosure, and exact de Casteljau subdivision. These are classical polynomial methods applied below to the source's rational derivative numerators. A failed sign test is inconclusive, not evidence of a root or a violation.

Volume 1's local differential representation assumes finite absolute moments and an integrable Taylor remainder on the actual interaction segments. It is exact on a finite polynomial input space, without asserting polynomial closure or an inverse. Its source-centered expansion differentiates the product of moment field and input; this is different from freezing the coefficient before differentiation. These qualifications preclude a shortcut from finite moment matching to a global signed determinant theorem.

3 Polynomial Rows and Signed Moment Spaces

For each integer $N \geq 1$ set

$$\begin{aligned} n = 48N, \quad a = 11N, \quad b = 7N, \quad q = g = 4N, \quad h = 2N, \\ L = 59N = n + a = n + b + q, \quad C = 63N, \quad H = 65N, \quad A = 19N. \end{aligned} \tag{3.1}$$

For $-1 < t < 1$, put $f(t) = \sqrt{1-t^2} > 0$ and $w = t/(1+f)$. The identities $t = 2w/(1+w^2)$ and $f = (1-w^2)/(1+w^2)$ identify the involution $w \mapsto w^{-1}$ with $f \mapsto -f$. For $0 \leq r < n$,

$$R_r = (1-t)^h t^{C-1} w^{r-g}, \qquad P_r = \frac{R_r + R_r^*}{2}, \qquad D_r = \frac{t(R_r^* - R_r)}{2f}. \tag{3.2}$$

The formula for R_r is initially a rational expression on the two-sheeted parametrization; the symmetric expressions are ordinary integer polynomials.

Proposition 3.1 (Exact row and contact identities)**.** *For every $N \geq 1$ and $0 \leq r < 48N$, with $d = |r-g|$,*

$$\begin{aligned} P_r &= (1-t)^h t^{C-1} T_d(1/t), \\ D_r &= \operatorname{sgn}(r-g)(1-t)^h t^{C-1} U_{d-1}(1/t), \end{aligned} \tag{3.3}$$

where $T_0 = 1, T_1 = x, U_{-1} = 0, U_0 = 1$ and both families satisfy $V_{j+1} = 2xV_j - V_{j-1}$. Their supports lie respectively in $[A, H-1] \cap \mathbb{Z}$ and $[A+1, H-1] \cap \mathbb{Z}$. As formal Taylor series at 0 on the branch $f(0) = 1$,

$$\frac{tP_r}{f} - D_r = \frac{tR_r}{f} = O(t^{C+r-g}) = O(t^L). \tag{3.4}$$

Proof. The Laurent identities $T_d((w+w^{-1})/2) = (w^d + w^{-d})/2$ and $U_{d-1}((w+w^{-1})/2) = (w^{-d} - w^d)/(w^{-1} - w)$, with $(w+w^{-1})/2 = 1/t$ and $(w^{-1}-w)/2 = f/t$, give (3.3), including $D_g = 0$. Since $d \leq n-1-g$, the least possible powers are $C - 1 - (n-1-g) = A$ and $A+1$; the greatest is $C - 1 + h = H - 1$. Integer Chebyshev coefficients prove integrality. Subtracting the two definitions gives $tP_r/f - D_r = tR_r/f$ exactly. Finally $w = t/2 + O(t^3)$ and $C + r - g \geq C - g = L$ give the Taylor order. No estimate at $t = \pm 1$ is used to infer this formal contact. □

For nonnegative integers i, j, define the absolutely convergent moments

$$\begin{aligned} M(i,j) &= \int_{-1}^{1} \int_{0}^{1} \frac{|t|}{\sqrt{1-t^2}} \frac{t^i s^j}{1-ts}\, ds\, dt, \\ Z(i,j) &= \int_{0}^{1} \int_{0}^{1} \frac{t^i s^j}{1-ts}\, ds\, dt. \end{aligned} \tag{3.5}$$

They extend bilinearly to polynomials. Indeed, integration in s of $(1-|t|s)^{-1}$ gives $-\log(1-|t|)/|t|$; after multiplying by $|t|/f(t)$ the remaining logarithmic endpoint singularity is integrable. Bounded polynomial factors cause no difficulty. This also supplies an absolute majorant for termwise geometric expansion, including negative t.

Set $c_l = 4^{-l}\binom{2l}{l}$ for integers $l \geq 0$ and $c_z = 0$ otherwise. Then

$$m_i = \int_{-1}^{1} t^i \frac{|t|}{f(t)}\, dt = \begin{cases} 2/((i+1)c_{i/2}), & i \text{ even}, \\ 0, & i \text{ odd}, \end{cases} \qquad m_{-1} = 0. \tag{3.6}$$

For even i, substituting $u = t^2$ reduces the integral to $\int_0^1 u^{i/2}(1-u)^{-1/2}\, du$; integration by parts gives $(i+1)m_i = i m_{i-2}$ and $m_0 = 2$. Consequently

$$M(i,j) = \sum_{u \geq 0} \frac{m_{i+u}}{j+u+1}, \qquad M(i,j) - M(i+1,j+1) = \frac{m_i}{j+1}. \tag{3.7}$$

3.1 Rational arrays and the two transcendental starts

Writing $K_d^- = M(d,0)$ and $K_d^+ = M(0,d)$, telescoping (3.7) with the moment recurrence gives

$$\begin{aligned} dK_d^- &= (d-1)K_{d-2}^- + m_{d-2} + m_{d-1} && (d \ge 2), \\ (d-1)K_d^+ &= (d-2)K_{d-2}^+ + 2/(d-1) && (d \ge 2), \\ K_0^- &= K_0^+ = 4G, \quad K_1^- = 2, \quad K_1^+ = \tfrac{3}{2}\zeta(2). \end{aligned} \tag{3.8}$$

For the first start, integration in s followed by $t = \sin\theta$ gives

$$M(0,0) = \int_0^{\pi/2} \log\frac{1+\sin\theta}{1-\sin\theta}\, d\theta = -4\int_0^{\pi/4} \log\tan v\, dv = 4G.$$

The last equality follows by $x = \tan v$ and the series of $(1+x^2)^{-1}$ integrated against $-\log x$. Also $M(0,1) = \sum_{k\ge1}(2k^2c_k)^{-1} = (\arcsin 1)^2 = \pi^2/4$: the Taylor coefficients follow by solving $(1-x^2)y'' - xy' = 2$, $y(0) = y'(0) = 0$; monotone convergence applies at 1. The classical identity $\zeta(2) = \pi^2/6$ gives (3.8).

Define $k_d^\pm$ by the same rational recurrences with $k_0^- = k_0^+ = k_1^+ = 0$, $k_1^- = 2$. Define

$$M^0(i,j) = \begin{cases} k_{i-j}^- - \displaystyle\sum_{k=1}^{j} \frac{m_{i-j+k-1}}{k}, & i \ge j, \\ k_{j-i}^+ - \displaystyle\sum_{k=j-i+1}^{j} \frac{m_{k-(j-i)-1}}{k}, & i < j. \end{cases} \tag{3.9}$$

If $B_i^{(d)} = \sum_{k=1}^{i} k^{-d}$ and $B_0^{(d)} = 0$, let

$$Z^0(i,j) = \begin{cases} -B_i^{(2)}, & i = j, \\ (B_i^{(1)} - B_j^{(1)})/(i-j), & i \ne j. \end{cases} \tag{3.10}$$

Propagation of the homogeneous boundary recurrences, followed by diagonal reduction, proves

$$\begin{aligned} M(i,j) &= M^0(i,j) + 4Gc_{(i-j)/2} + \tfrac{3}{2}\zeta(2)c_{(j-i-1)/2}, \\ Z(i,j) &= Z^0(i,j) + [i=j]\zeta(2). \end{aligned} \tag{3.11}$$

For Z, this also follows directly by partial fractions of $\sum_{u\ge0}((i+u+1)(j+u+1))^{-1}$. The sign in the off-diagonal formula is positive, for example $Z^0(0,1) = 1$; the diagonal rational part is negative, for example $Z^0(1,1) = -1$.

Corollary 3.2 (Contact cancels exactly one extra period)**.** *For $b \le j < L$, every raw entry $F_j(r) = M(P_r, s^j) - \frac{3}{2}Z(D_r, s^j)$ lies in $\mathbb{Q} + \mathbb{Q}G$. The determinant*

$$\Delta_N = \det_{0\le r,k<n}\left[M(P_r, s^{b+k}(1-s)^q) - \frac{3}{2}Z(D_r, s^{b+k}(1-s)^q)\right] \tag{3.12}$$

therefore lies in $\mathbb{Q}[G]$, and is rational if $G \in \mathbb{Q}$.

Proof. The coefficient of $\zeta(2)$ is

$$\frac{3}{2}\left(\sum_i [t^i]P_r c_{(j-i-1)/2} - [t^j]D_r\right) = \frac{3}{2}[t^j](tP_r/f - D_r) = 0.$$

Expanding $(1-s)^q$ uses the integer matrix $\mathcal{T}_{j,k} = (-1)^{j-b-k}\binom{q}{j-b-k}$, $b \le j < L$, $0 \le k < n$. The last raw degree is $b+n-1+q = L-1$, so all columns remain inside the exact contact range. Determinant expansion gives $\mathbb{Q}[G]$ of degree at most n, not generally a linear form in G. Rationality does not imply nonzero value. □

3.2 The actual finite representation

Use the physical space

$$\mathcal{U}_N = \operatorname{span}_{\mathbb{Q}}\{(P_r, D_r) : 0 \le r < n\} \subset \mathbb{Q}[t]_{<H} \oplus \mathbb{Q}[t]_{<H},$$

with the functionals $\ell_j(P, D) = M(P, s^j) - \frac{3}{2}Z(D, s^j)$ and the column space $\mathcal{V}_N = s^b(1-s)^q\mathbb{Q}[s]_{<n}$. The row pairs are linearly independent: if $\sum u_r(P_r, D_r) = 0$, then $\sum u_r R_r = 0$ by $R_r = P_r - fD_r/t$. Dividing its nonzero common factor gives $\sum u_r w^r = 0$ on an interval, hence all $u_r = 0$. Thus $\dim\mathcal{U}_N = n$, but the n filtered functionals need not be unisolvent merely because the row pairs form a basis. That is exactly the missing rank hypothesis in a naive application of the finite Ram map.

The matrix is mixed and signed, not a positive Gram matrix. Even two positive matrices B, C can have singular difference $B - C$; taking $B = C = I$ proves the non-implication. Here the row and column polynomial spaces are different and the Z term is subtracted. Neither the positivity argument of TR250 nor a positive-kernel moment matrix from Volume 1 establishes $\Delta_N \neq 0$.

4 Arithmetic Chain: Source Claims and Exact Dependencies

This section describes all arithmetic source components from their actual proofs. A separate independent exact replay and bounded proof review supports this part of the audit; its checked scope and limitations are reported in Section A.

4.1 The 2-adic smoothing and alternation

Under the single hypothesis $G \in \mathbb{Q}$, embed it in $\mathbb{Q}_2$. The source constructs $M_{\mathrm{s}}(i, j) = \sum_{k\ge 0} m_{i+k}/(j+k+1)$ in $\mathbb{Q}_2$. For even u, $v_2(m_u) = 1 + u - v_2\binom{u}{u/2}$, so the terms tend to zero 2-adically and for $0 \le i, j < H$ have valuation at least $i - 2\log_2 H - 1$. There are two homogeneous discrepancies $e_1 = 4G - M_{\mathrm{s}}(0,0)$ and $e_2 = -M_{\mathrm{s}}(0,1)$, fixed independently of N. After contact cancellation each raw column splits into a smoothed column, a c-kernel column, and a D-coefficient column.

Writing unconvolved coefficients as $P_r/(1-t)^h = \sum_u p_{r,u}t^{C-1-u}$ and similarly for $d_{r,u}$, the Chebyshev recurrence gives $v_2(p_{r,u}), v_2(d_{r,u}) \ge u-1$. Cauchy–Binet must be applied before forgetting the distinct raw indices. Among m exceptional P columns the indices u are distinct, as are those of l exceptional D columns; repetitions within either group annihilate the determinant. No cross-group distinctness is needed. This produces the source lower estimate

$$v_2(\Delta_N) \ge -O_G(n\log(n+2)) + \min_{m,l\ge 0,\ m+l\le n}\big\{-(H-b)m + \tfrac{3}{2}m^2 + \tfrac{1}{2}l^2 + C(n-m-l)\big\}. \tag{4.1}$$

The minimization itself is elementary: $C \ge n$ makes the expression decrease in l up to $n - m$; at that boundary, with $\delta = 5/24$, it is

$$2\left(m - \frac{(2+\delta)n}{4}\right)^2 - \left(\frac{\delta}{2} + \frac{\delta^2}{8}\right)n^2.$$

Thus the asserted 2-adic cost is $505n^2/4608 + O_G(n\log n)$. The finite minimization is not a replacement for the smoothing and column-valuation hypotheses that precede it.

4.2 Odd primes require two denominator layers

For $2\sqrt{H} < p \le H$, excluding the fixed denominator primes of G, set $E(t) = (1-t^2)^{(p-1)/2} = \sum E_d t^d \in \mathbb{F}_p[t]$. For $0 \le i, j < H$ let $\ell = j \bmod p$, $d = (j-i-1) \bmod p$, $j' = \lfloor j/p \rfloor$, $i' = (i+1+d-\ell)/p - 1 \ge -1$. The source proves locally integral reductions

$$\begin{aligned} p^2 M^0(i,j) &\equiv E_d M^0(i',j') \pmod p, \\ p^2 Z^0(i,j) &\equiv [i \equiv j \pmod p]\, Z^0(\lfloor i/p \rfloor, j') \pmod p, \end{aligned} \tag{4.2}$$

where $M^0(-1,j') = k^+_{j'+1}$ is an algebraic convention, not an integral. The parity/carry proof uses $H < p^2/4$; this restriction cannot be dropped.

Coefficient extraction uses different shifts: $P'_u = [t^{(u+1)p+\ell}]tPE$, $D'_u = [t^{up+\ell}]D$. At low reduced degrees define $V_\ell = 2P'_1 - 3D'_1/2$ and $U_\ell = 2P'_{-1} - 3D'_0/2$. Then $p^2 F_\ell \equiv V_\ell$ and $p^2 F_{p+\ell} \equiv U_\ell - V_\ell$; $U_\ell = 0$ for $\ell \le A$. Pairs give an improvement from p^{-2} to p^{-1}, not to integrality. For $H/2 < p \le H$ in the large-prime range and with $p \nmid \operatorname{den}(G)$, set $J = H - p$, $K = L - p$. The central columns $J \le j < p$ in the actual raw range satisfy

$$pF_j \equiv -\sum_{0 \le \ell < J} \frac{V_\ell}{j-\ell} \pmod p.$$

The denominators are units. Retained noncentral columns furnish representatives of V_ℓ; integral shears by p times those columns cancel the represented residues. The unknown next layer of a retained p^{-2} column becomes integral after that multiplication, which is why the shear is legitimate. Residual rank then leaves at most

$$\begin{aligned} R &= K_+ + (K-A)_+ + (J - \max(b,K))_+, \\ S &= (\min(A,K) - b)_+ + (\min(b,J) - \max(0,K))_+ \end{aligned}$$

columns of possible orders p^{-2} and p^{-1} respectively. Integral inverse column changes and Cauchy–Binet yield the asserted $v_p(\Delta_N) \ge -\min(2n, n+R, 2R+S)$ at these permitted primes. The restriction $p \nmid \operatorname{den}(G)$ is a standing hypothesis of both local reductions, not an all-N conclusion about arbitrary primes. For fixed rational G, its finitely many denominator primes are eventually below $2\sqrt{65N}$ and their cost is included in the small-prime bound.

The continuous loss profile $d(x)$, $x = p/N$, is

range	$d(x)$	range	$d(x)$
$[0, 25]$	96	$[25, 29]$	$146 - 2x$
$[29, 65/2]$	88	$[65/2, 69/2]$	$153 - 2x$
$[69/2, 40]$	$222 - 4x$	$[40, 58]$	$182 - 3x$
$[58, 59]$	$124 - 2x$	$[59, 65]$	$65 - x$

Its area is $8609/2$. The prime number theorem, applied by upper and lower step functions on $[0, 65]$, gives $N^{-1} \sum_{p \le 65N} d(p/N) \log p \to \int d$. The small-prime cost $O(n\sqrt{H} \log H + n \log \operatorname{den} G)$ is $o(n^2)$; primes $p > H$ are eventually integral. For nonzero rational Δ_N, the product formula therefore gives the asserted lower bound

$$\liminf \left(n^{-2} \log |\Delta_N| - \tfrac{1}{2} \log 2 \right) \ge -\frac{8609}{4608} - \frac{2809}{4608} \log 2 > -2.29084. \tag{4.3}$$

Its quantifier is along any sequence of nonzero determinants with $N \to \infty$ under the fixed rationality hypothesis, not all real numbers substituted for G uniformly with one error constant.

4.3 Nonvanishing is a different finite-to-infinite bridge

At scale $N = p > 260$, the source uses a p-dimensional space of palindromic polynomials of degree at most $2p-2$ over $\mathbb{F}_p$. The bases

$$U_\ell = (2w)^\ell (1+w^2)^{p-1-\ell}, \qquad Q_i = w^i \sum_{j=0}^{p-i-1} w^{2j} \quad (0 \le \ell, i < p)$$

have triangular transition matrix $\mathbf{a}$ with diagonal 2^{-i}. Writing $Q'_0 = 0$, $Q'_i = Q_{p-i}$ gives $\mathbf{b} = \mathbf{a}\Pi$. Frobenius reduces the rows to base scale 1 and one successor row. The auxiliary row 48 has minimum degrees 18 and 19 in its P and D components and must not be truncated at the usual $19, 20$.

Let $\mathcal{B}$ be the rational 49×48 matrix obtained by the fourth forward difference of $M^0(\underline{P}_r, 7+k) - 3Z^0(\underline{D}_r, 7+k)/2$. Let $\mathcal{B}_0$ use rows 0–47 and $\mathcal{B}_1$ rows 1–48. The source supplies 48 nonzero pivots modulo 101 for each of $\mathcal{B}_0$, $\mathcal{B}_0 + \mathcal{B}_1$, and $\mathcal{B}_0 - \mathcal{B}_1$. Only the middle calculation needs swaps, at $(30, 31)$ and $(45, 46)$ with zero-based indices. The separate exact replay gives determinants $27, 50, 74$ modulo 101 in this order and agrees with every one of the 144 displayed pivots.

In the stated block order the reduced growing matrix is $\mathcal{L}_p = \mathbf{a}^T \otimes \mathcal{B}_0 + \mathbf{b}^T \otimes \mathcal{B}_1$. Right multiplication by $(\mathbf{a}^T)^{-1} \otimes I$ gives $I \otimes \mathcal{B}_0 + \Pi^T \otimes \mathcal{B}_1$. The eigenvalues of Π are $0, 1, -1$ with multiplicities $1, (p-1)/2, (p-1)/2$. Hence

$$\begin{aligned} \det \mathcal{L}_p = (\det \mathbf{a})^{48} \det \mathcal{B}_0 \det(\mathcal{B}_0 + \mathcal{B}_1)^{(p-1)/2} \\ \cdot \det(\mathcal{B}_0 - \mathcal{B}_1)^{(p-1)/2}. \end{aligned} \tag{4.4}$$

If the digit reductions and three fixed nonzero determinants hold, only a fixed finite set of primes is excluded, and $\det(p^2 \mathcal{F}_p)$ is a unit. Thus $v_p(\Delta_p) = -96p$ for every sufficiently large prime p. A successful check modulo 101 of fixed rational matrices does not alone prove this varying-prime bridge.

5 Real-Place Geometry and Uniform Evaluation

Put $\alpha = 11/48$, $\beta = 7/48$, $\gamma = 1/12$, $\eta = 1/24$, and $D_* = n - 1 - 2g = 40N - 1$. Let $V(y) = \prod_{i<j}(y_j - y_i)$, $\mathcal{V}(y) = |V(y)|$, and $d\mu(t) = |t|\, dt/f(t)$. The row functions

$$A_r(t) = P_r(t) - \tfrac{3}{2}\mathbf{1}_{t>0}\frac{f(t)}{t} D_r(t), \qquad \psi_k(s) = s^{b+k}(1-s)^q$$

are bounded for each fixed N. Double determinant integration gives

$$\Delta_N = \frac{1}{(n!)^2} \int\int \det[A_r(t_i)] \det[(1 - t_i s_j)^{-1}] \det[\psi_k(s_j)]\, ds\, d\mu^n(t), \tag{5.1}$$

on $(-1, 1)^n \times (0, 1)^n$. This is the classical Andréief identity, obtained by expanding and relabeling permutations; each integration list contributes one factor $n!$. The scalar majorant used in (3.5) justifies every finite permutation expansion and Fubini. The multiplicative Cauchy determinant is $V(t)V(s)/\prod_{i,j}(1 - t_i s_j)$; clearing denominators gives a separately alternating polynomial with constant quotient by $V(t)V(s)$, and expansion at $s = 0$ fixes that constant as 1.

Under $t_i = 2x_i/(1 + x_i^2)$, $d\mu(t_i) = 4|x_i|\, dx_i/(1 + x_i^2)^2$. The two sheet coefficients (ξ_f, ξ_n) are $(1/2, 1/2)$ for $x < 0$ and $(-1/4, 5/4)$ for $x > 0$. Thus

$$\mathcal{R}_n(x) = \det[\xi_f(x_i) x_i^{g-r} + \xi_n(x_i) x_i^{r-g}] \tag{5.2}$$

is the row determinant after extracting $\prod_i t_i^{C-1}(1 - t_i)^h$. The signs are essential: on $t > 0$, $A_r = (5R_r - R_r^*)/4$.

5.1 Why clustered nodes do not require an inverse Vandermonde

For distinct nonzero $x_i \in (-1,1)$, split into the cases

$$\Omega_{2,n}: \quad \sum_i \frac{1-x_i^2}{1+x_i^2} \le D_*; \qquad \Omega_{1,n}: \quad \text{the strict reverse inequality.} \tag{5.3}$$

Factoring the far-sheet weight leaves $x_i^{n-1-r} + c_i x_i^{D_*} x_i^r$, where $c_i = 1$ for $x_i < 0$ and $c_i = -5$ for $x_i > 0$. The source's interpolation lemma constructs a holomorphic h_* near the closed disk with $h_*(x_i) = c_i x_i^{D_*}$ and $\|h_*\|_\infty \le K_0 n$, $K_0 = 10e^{12}$, in case 2. Here are its quantitative ingredients, since uniformity in colliding nodes is indispensable.

Set $B(z) = \prod_i (z - x_i)/(1 - x_i z)$ and $F(z) = z^{D_*}/B(z)$. For $0 < u \le 1$,

$$\frac{d}{d\log u} \log \left| \frac{iu - x}{1 - ixu} \right| = \frac{(1-x^4)u^2}{(u^2+x^2)(1+x^2u^2)} \le \frac{1-x^2}{1+x^2}.$$

Integration to 1 and (5.3) give $|B(iu)| \ge u^{D_*}$. For $1 \le |u| \le 1 + 2/n$, each Blaschke factor has modulus at least 1, so $|F(iu)| \le e^2$ on the extended diameter Γ. Define

$$\mathcal{C}(z) = \frac{1}{2\pi i} \int_\Gamma \frac{6F(w)}{w - z}\, dw, \qquad h_*(z) = c_\pm z^{D_*} - B(z)\mathcal{C}(z).$$

The left-minus-right jump is $6F$, precisely cancelling $(c_- - c_+)z^{D_*}$. The resulting function glues holomorphically; its exterior neighborhood may depend on the nodes. The poles $1/x_i$ are outside the closed disk. At x_i, B vanishes and the prescribed interpolation values follow.

Near $\pm i$, $|F'/F| < 4n$ on radius-1/4 disks; thus $|F| \le e^{12}$ within distance $3/n$ of those points. On the unit circle at horizontal distance at least $1/(10n)$ from the diameter, its Cauchy integral is bounded by $30e^2 n$. Otherwise deform just the adjacent end of the diameter into the opposite half-plane by a rectangle of horizontal width $1/n$ and vertical span $3/n$. The contour stays within $3/n$ of $\pm i$ there, has total length at most 3, and is at least $1/(2n)$ from z. It does not cross z or any real node, hence the value is unchanged and bounded by $6e^{12}n$. Together with $|B| = 1$ on the circle and the maximum modulus principle these bounds give $K_0 n$. There is no separation constant for the nodes in this argument.

For $Q(z) = \prod_i (1 - x_i z)$ use

$$\mathcal{H}_Q = \{v/Q : v \in \mathbb{C}[z], \deg v < n\} \subset H^2(\mathbb{D}), \quad \|u\|^2 = \frac{1}{2\pi} \int_0^{2\pi} |u(e^{i\theta})|^2\, d\theta.$$

The reversal $R(v/Q) = z^{n-1} v(1/z)/Q$ is a complex-linear isometry, using the real coefficients of Q. The independent reproducing kernels $(1 - x_i z)^{-1}$ belong to this n-dimensional space, so orthogonal projection Π_Q preserves evaluations at the x_i. The compression $J = \Pi_Q M_{h_*}|_{\mathcal{H}_Q}$ satisfies $\|J\| \le K_0 n$. If $E(v/Q) = (v(x_i))_i$, then

$$E(I + JR)R(v/Q) = (\widetilde{v}(x_i) + h_*(x_i)v(x_i))_i.$$

Taking determinants cancels $\det E = V(x)$ algebraically. Consequently

$$|\mathcal{R}_n(x)| \le (1 + K_0 n)^n \mathcal{V}(1/x) \prod_i |x_i|^g. \tag{5.4}$$

No estimate of E^{-1} is used. Small node separations therefore do not introduce an omitted large factor.

On the other hand, expand (5.2) over all choices $z_i \in \{x_i, 1/x_i\}$ and put $z_i = \tan\phi_i$. The sine Vandermonde is $2^{-\binom{n}{2}}$ times a unit-circle Vandermonde, whose monomial matrix has column norm $\sqrt{n}$. Hadamard gives

$$|\mathcal{R}_n(x)| \le (3/2)^n n^{n/2} 2^{-\binom{n}{2}} \prod_i (1+x_i^2)^{(n-1)/2} |x_i|^{g-(n-1)}. \tag{5.5}$$

Indeed the far-to-near weight ratio is $|x_i|^{-D_*} \ge 1$, and the sum of absolute sheet weights is at most $(3/2)^n$. This bound holds in both cases.

5.2 Exact majorants and measure accounting

Define

$$\begin{aligned}
\mathcal{J}_n &= \frac{\mathcal{V}(t)\mathcal{V}(s)^2}{\prod_{i,j}(1-t_i s_j)} \prod_j s_j^b (1-s_j)^q \prod_i |t_i|^{C-1}(1-t_i)^h, \\
\mathcal{I}_{2,n} &= \mathcal{J}_n \mathcal{V}(1/x) \prod_i |x_i|^g, \\
\mathcal{I}_{1,n} &= \mathcal{J}_n 2^{-\binom{n}{2}} \prod_i (1+x_i^2)^{(n-1)/2} |x_i|^{g-(n-1)}.
\end{aligned}$$

Both are nonnegative on the configurations just specified, with $s_j \in (0,1)$. Since $\mu((-1,1)) = 2$, (5.1)–(5.5) give

$$\frac{\log|\Delta_N|}{n^2} - \frac{1}{2}\log 2 \le \max_{\kappa=1,2} \sup_{\Omega_{\kappa,n}} \left(\frac{\log \mathcal{I}_{\kappa,n}}{n^2} - \frac{1}{2}\log 2 \right) + O\left(\frac{\log(n+2)}{n} \right). \tag{5.6}$$

The error accounts for $(n!)^{-2}$, mass 2^n, and both prefactors. Repeated s or nonzero x nodes make the original relevant determinant zero. Zero and boundary nodes form null sets; their approaches remain in the suprema and cannot be excluded by an unproved separation hypothesis.

6 Uniform Energy Reduction With Endpoints Retained

For real exponentially decaying sequences $u = (u_k)_{k\ge1}$, meaning $|u_k| \le Kr^k$ for some $K < \infty$, $r < 1$, put

$$\|u\|_*^2 = \sum_{k\ge1} \frac{u_k^2}{k}, \qquad T(u,x) = \sum_{k\ge1} \frac{u_k T_k(x)}{k}, \qquad S(u,x) = \sum_{k\ge1} \frac{u_k x^k}{k}.$$

These series are uniformly convergent for $|x| \le 1$. Define

$$\begin{aligned}
W_\kappa(x) &= \frac{19}{48}\log|x| + \frac{1}{12}\log(1-x) - \left(\frac{\kappa}{2} + \frac{17}{48}\right)\log(1+x^2), \\
W_s(s) &= \frac{7}{48}\log s + \frac{1}{12}\log(1-s), \qquad D(x) = \frac{1}{6} - \frac{2x^2}{1+x^2}.
\end{aligned} \tag{6.1}$$

The scalar $D(x)$ is unrelated to the row polynomial $D_r(t)$. For trials p, v and $\lambda_2 = 0$, $\lambda_1 \ge 0$, let

$$\begin{aligned}
X_\kappa(x) &= W_\kappa(x) + \lambda_\kappa D(x) - 2\kappa T(p,x) - S(v,x), \\
Y_\kappa(s) &= W_s(s) + 2T(v,s), \\
\mathcal{B}_\kappa(p,v,\lambda_\kappa) &= -\frac{11}{16}\log 2 + \kappa\|p\|_*^2 + \tfrac{1}{2}\|v\|_*^2 + \sup_{[-1,1)\setminus\{0\}} X_\kappa + \sup_{(0,1)} Y_\kappa.
\end{aligned} \tag{6.2}$$

Proposition 6.1 (Source energy estimate, rederived)**.** *For every such fixed pair of trials in each case,*

$$\limsup_{N\to\infty}\ \sup_{\Omega_{\kappa,n}} \left(n^{-2}\log \mathcal{I}_{\kappa,n} - \tfrac{1}{2}\log 2\right) \le \mathcal{B}_\kappa.$$

Combined with (5.6)*, the maximum of the two bounds is an upper bound on the corresponding normalized determinant limsup.*

Proof. Write $\langle F(x)\rangle = n^{-1}\sum_i F(x_i)$; double averages divide by n^2, and subscript $\neq$ omits equal-index pairs. The elementary substitution identities

$$|t-t'| = \frac{2|x-x'|(1-xx')}{(1+x^2)(1+x'^2)}, \quad 1-ts = \frac{1-2xs+x^2}{1+x^2}, \quad 1-t = \frac{(1-x)^2}{1+x^2}$$

give the exact finite identity

$$\begin{aligned} n^{-2}\log \mathcal{I}_{\kappa,n} - \tfrac{1}{2}\log 2 = c_{\kappa,n}\log 2 + \langle W_\kappa(x)\rangle + \langle W_s(s)\rangle + \frac{\kappa+2}{2n}\langle \log(1+x^2)\rangle \\ + \frac{\kappa}{2}\langle \log|x-x'|\rangle_{\neq} + \frac{1}{2}\langle \log(1-xx')\rangle_{\neq} \\ + \langle \log|s-s'|\rangle_{\neq} - \langle \log(1-2xs+x^2)\rangle, \qquad (6.3) \\ c_{\kappa,n} = C/n + (\kappa-2)/2 - (\kappa+1)/(2n). \end{aligned}$$

In particular the total $|x_i|$ exponent is $(C-1)+g-(n-1) = a+2g$ exactly; there is no hidden $n^{-1}\log|x|$.

Let $u=\cos\theta$, $u'=\cos\theta'$ and, for $0\le r<1$, define

$$L_r(u,u') = -\log 2 + \log|1-re^{i(\theta+\theta')}| + \log|1-re^{i(\theta-\theta')}| = -\log 2 - 2\sum_{k\ge 1}\frac{r^k T_k(u)T_k(u')}{k}.$$

The last equality follows from the logarithmic power series and cosine addition. At distinct u,u', $L_1 = \log|u-u'|$. The other kernels are $\log(1-zz') = -\sum z^k z'^k/k$ and $-\log(1-2zs+z^2) = 2\sum z^k T_k(s)/k$ for $|z|,|z'|<1$. Direct summation of empirical diagonals at $r=1$ is invalid.

Fix $0<\varepsilon<1/8$, and damp with $\tau = 1-\varepsilon$, $\rho(x) = 1-\varepsilon(1-x)$, $\sigma(s) = 1-\varepsilon\sqrt{1-s}$. Use L_{τ^2} for the x cosine kernel, $L_{\sigma(s)\sigma(s')}$ for the s kernel, $\log(1-\rho(x)\rho(x')xx')$ for the power kernel, and $-\log|1-\rho(x)\sigma(s)xe^{i\arccos s}|^2$ for the cross kernel. Each original kernel is at most the new one plus $O(\varepsilon)$, uniformly in all nodes. For cosine kernels use $|1-r\zeta|^2 - r|1-\zeta|^2 = (1-r)^2$. For $xx'<0$, the power-kernel difference is at most $1-\rho(x)\rho(x') \le 4\varepsilon$; for $xx'\ge 0$ it is nonpositive. For the cross kernel, $z = 1-xe^{i\theta}$ obeys $|z_\varepsilon - z| \le \varepsilon|x|(1-x+\sqrt{1-s}) \le 3\varepsilon|z|$. The last bound follows separately on $x\le 0$, $0\le x\le 1/2$, and $x\ge 1/2$, using $|z|^2 = (1-x)^2 + 2x(1-s)$ in the last range. Thus the logarithmic error is at most 6ε in the required upper-bound direction.

Set $A_k = \langle \tau^k T_k(x)\rangle$, $B_k = \langle \rho(x)^k x^k\rangle$, $C_k = \langle \sigma(s)^k T_k(s)\rangle$. Including diagonals gives exactly

$$-\frac{\kappa}{2}\log 2 - \kappa\sum_k A_k^2/k, \qquad -\log 2 - \tfrac{1}{2}\sum_k (B_k - 2C_k)^2/k$$

for the two groups of interactions. Matching the damping of the cross and pure terms is what makes the second identity true. Together with $c_{\kappa,n}$, the constant becomes $(-11/16 - (\kappa+1)/(2n))\log 2$.

Removing the added x cosine diagonals costs $O_\varepsilon(1/n)$. The power diagonals satisfy $1-\rho(x)^2x^2 \ge 1-x$ for $x\ge 0$ and are at least ε for $x\le 0$. Their cost is at most $O_\varepsilon(1/n) - (2n)^{-1}\langle\log(1-x)\rangle$. Since $1-\sigma(s)^2 \ge \varepsilon\sqrt{1-s}$, the s cosine cost is at most $O_\varepsilon(1/n) - n^{-1}\langle\log(1-s)\rangle$. The cross average already contains all pairs.

Apply $-a^2 \le b^2 - 2ab$ with $b = p_k$ and $b = v_k$ respectively. In case 1, (5.3) implies $\langle D \rangle > -1/n$, so adding $\lambda_1 \langle D \rangle$ costs at most λ_1/n; this fixes both its sign and the requirement $\lambda_1 \ge 0$. We are left with damped versions of X_κ, Y_κ whose endpoint log coefficients are $1/12 - 1/(2n)$ and $1/12 - 1/n$. For $n \ge 24$ both are at least $1/24$.

If $P_1 = \sum |p_k|$, $V_1 = \sum |v_k|$, then $1 - (1-d)^k \le kd$ gives uniform tangent errors at most $2\kappa\varepsilon P_1 + 2\varepsilon V_1$ and $2\varepsilon V_1$. All other terms are uniformly bounded except the displayed endpoint logs. The positive log coefficients force decay to $-\infty$ at $x = 0, 1$ and $s = 0, 1$, uniformly for $n \ge 24$ and small ε. Comparison values at $x = -1/2$, $s = 1/2$ give fixed finite lower bounds. Therefore maximizers stay in fixed compact subsets, with the regular endpoint $x = -1$ retained. On these sets take $n \to \infty$ first with ε fixed, removing $O_\varepsilon(1/n)$; then take $\varepsilon \downarrow 0$. This order proves the asserted uniform supremum bound without claiming uniform convergence of the raw singular square series. □

7 Exact Trial Data and Rational Barrier Replay

All terminating decimals in this section are exact rational numbers. The source trials have the form $u_k = l_k + \sum_z r_z z^k$, with $l_k = 0$ for $k > d$. Use $d = 10$, $\lambda_2 = 0$ for case 2 and $d = 8$, $\lambda_1 = 2.47405979$ for case 1. The finite coefficients below are multiplied by 10^{-8} and read across successive rows of each sequence.

κ	u	finite coefficients				
2	p	45559127	-50750856	-6578767	13970217	4786184
		-4292433	-576704	1311615	671564	-346453
2	v	-23910158	21152432	-2885110	-11558199	6485289
		1456821	-1912176	-2263524	2742210	-1162454

κ	u	finite coefficients			
1	p	11913521	-79993701	-21956443	37903579
		10744022	-2073193	570246	-24939103
1	v	-89913025	52874280	45168341	-30708629
		-19269841	33922112	9819141	-16992389

Case 1 has no tails. For case 2 the tail data are:

u	z	$10^8 a$	$10^8 b$
p	.85	-9338452	0
	.94	-2141509	0
	$.7i$	-66277922	-31907569
	$.85i$	-1231651	6002645
	$.092 + .92i$	-3225918	8928234
	$-.092 + .92i$	-2105536	-9091287
v	$-.8$	15199211	0
	$-.96$	4451662	0
	.88	2545398	0
	.95	-4932634	0
	.984	11618157	0
	$.78i$	27238714	-38447936
	$.9i$	-31341084	-30188786
	$.955i$	-6693542	11912254
	$.984i$	2055213	-21715849

For real z use $r_z = a$ once. For nonreal z use $r_z = (a - ib)/2$, $r_{\bar{z}} = \overline{r_z}$, so the contribution is $a\Re(z^k) + b\Im(z^k)$. There are 10 and 13 tail terms after conjugates are counted; their base moduli are at most .94 and .984 respectively, and all $|l_k|, |r_z| < 1$. These facts are exact rational inequalities after squaring the complex moduli.

7.1 Finite logarithm formulas and branch control

Expanding $-\operatorname{Log}(1-z) = \sum_{k\ge1} z^k/k$ gives

$$\begin{aligned} \|u\|_*^2 &= \sum_{k=1}^{d} \frac{l_k^2 + 2l_k \sum_z r_z z^k}{k} - \sum_{z,z'} r_z r_{z'} \operatorname{Log}(1 - zz'), \\ T(u,x) &= \sum_{k=1}^{d} \frac{l_k T_k(x)}{k} - \tfrac{1}{2} \sum_z r_z \operatorname{Log}(1 - 2xz + z^2), \\ S(u,x) &= \sum_{k=1}^{d} \frac{l_k x^k}{k} - \sum_z r_z \operatorname{Log}(1 - xz). \end{aligned} \tag{7.1}$$

Conjugation makes each expression real. The norm and power log arguments have positive real part. For $x = \cos\theta$, $1 - 2xz + z^2 = (1 - ze^{i\theta})(1 - ze^{-i\theta})$; both factors have positive real part, so their arguments sum strictly inside $(-\pi, \pi)$. Thus their log sum is the principal logarithm of the product, even when the product has negative real part. This avoids silently dropping a multiple of $2\pi i$ with complex weights.

For a positive rational $s = 2^m y$, $1 \le y \le 2$, put

$$H(y) = 2\sum_{j=0}^{17} \frac{q^{2j+1}}{2j+1}, \quad q = \frac{y-1}{y+1}, \qquad 0 \le \log y - H(y) \le \frac{2q^{37}}{37(1-q^2)}.$$

The interval for $m\log 2 + \log y$ uses the sign of m, not an unsigned error subtraction. For complex $a + ib \ne 0$, its real log part is $\frac{1}{2}\log(a^2 + b^2)$. Rotate to a principal octant so the residual ratio t is rational and $|t| \le 1/2$; rotations can use $(a, b) \mapsto (a + b, b - a)$ or its inverse up to positive scaling. With

$$J(t) = \sum_{j=0}^{23} \frac{(-1)^j t^{2j+1}}{2j+1}, \qquad |\arctan t - J(t)| \le |t|^{49}/49,$$

and $\pi/4 = \arctan(1/2) + \arctan(1/3)$, the principal argument is enclosed by $k(J(1/2)+J(1/3))+J(t)$ with the corresponding signed errors. On the negative real axis use π. All inputs here lie in $[2^{-100}, 2^{100}]$; denominator factors have modulus at least $(.016)^2$ and at most $(1 + .984)^2$, and real endpoint arguments are at least .0007.

The author implementation uses exact rational operations followed by outward rounding to multiples of 10^{-45} after scalar interval operations. Series remainders are included before rounding. The maximum log error estimates, with $h_0 = 2(1/3)^{37}/(37(1 - 1/9))$ and $j_0 = (1/2)^{49}/49$, are $101h_0$ for real logs and $2 \cdot 101h_0 + 9j_0 < 7 \cdot 10^{-16}$ as a conservative complex-log allowance. The total log coefficient masses are at most 284.5 for the weighted norm, 35 for X, and 14 for Y. The exact interval calculation uses the individual remainders, rather than only these coarse mass bounds. Decimal diagnostic displays are never used for certification.

7.2 Derivatives and complete rational numerator specification

Set

$$t_u(x) = \sum_{k=1}^{d} l_k U_{k-1}(x) + \sum_z \frac{r_z z}{1 - 2xz + z^2}, \qquad h_u(x) = \sum_{k=1}^{d} l_k x^{k-1} + \sum_z \frac{r_z z}{1 - xz}.$$

Then

$$\begin{aligned} X'_\kappa(x) &= \frac{19}{48x} - \frac{1}{12(1-x)} - \left(\kappa + \frac{17}{24}\right)\frac{x}{1+x^2} - \frac{4\lambda_\kappa x}{(1+x^2)^2} - 2\kappa t_p(x) - h_v(x), \\ Y'_\kappa(x) &= \frac{7}{48x} - \frac{1}{12(1-x)} + 2t_v(x). \end{aligned} \tag{7.2}$$

Use the exact denominators

$$\begin{aligned} Q_X &= x(1-x)(1+x^2)^{3-\kappa}\prod_{z\in p}(1-2xz+z^2)\prod_{z\in v}(1-xz), \\ Q_Y &= x(1-x)\prod_{z\in v}(1-2xz+z^2), \qquad A_X = Q_X X'_\kappa, \quad A_Y = Q_Y Y'_\kappa. \end{aligned} \tag{7.3}$$

Together with the tables, these specify every rational coefficient of the four numerator polynomials, without fitting or root approximation. Conjugate factors have positive products; real-base factors are positive on $[-1,1]$. Thus Q_X has the sign of x, and $Q_Y > 0$ on $(0,1)$. The author verifies that the degrees are $36, 24, 13, 9$ in order X_2, Y_2, X_1, Y_1. Positive denominator clearing and removal of positive rational content yield the primitive integer numerators saved in evidence.

The open root brackets are $(m, m+2)/10^{10}$ for the following integers:

κ	function	m		
2	X	-9601109148	-8942317572	-7608305633
		-6503394794	-5185864065	-4015634158
		-3108806646	-2067921826	-1589849496
		1531948062	2072448179	3208186484
		4381119427	5851354199	7269030693
		8390277402	9332614564	9709786219
2	Y	176402802	330649406	764952882
		1227250753	2149465998	3048189112
		4322699096	5564757994	6801929373
		8031988371	8877037851	9577761832
		9838463999	9972727815	9992037196
1	X	-9917299785	-2259572153	2543808026
		4437270259	6348298970	
1	Y	532669786	2504239325	5701738806
		7966939383	9454490138	

Exact endpoint products $A(m/10^{10})A((m+2)/10^{10}) < 0$ hold for all 43 brackets. The source's Descartes transforms on the division intervals replay as follows:

κ	function	degree	division points	variations
2	X	36	$-1, 0, 1$	$9, 9$
2	Y	24	$0, 1/4, 1/2, 3/4, 1$	$5, 2, 2, 6$
1	X	13	$-1, -1/2, 0, 1/2, 1$	$1, 1, 2, 1$
1	Y	9	$0, 1$	5

The transform is $(1+t)^d A((a+bt)/(1+t))$, $t>0$. Its variations bound roots with multiplicity on (a,b). Since the exhibited sign-changing brackets already attain the bound on every such interval, there is exactly one simple root in each and no additional open-interval roots. All finite-valued division points are included among the value candidates.

8 Bernstein Exhaustion and a Sharper Bounded Certificate

Lemma 8.1 (Exact Bernstein enclosure)**.** *If* $P(a+(b-a)s)=\sum_{k=0}^{d}\alpha_k s^k$, $0\le s\le 1$, *then*

$$P(a+(b-a)s)=\sum_{j=0}^{d}\beta_j\binom{d}{j}s^j(1-s)^{d-j},\quad \beta_j=\sum_{k=0}^{j}\alpha_k\frac{\binom{j}{k}}{\binom{d}{k}}.$$

Thus every value is between $\min\beta_j$ *and* $\max\beta_j$. *Strict same-sign coefficients exclude zeros on the entire closed interval.*

Proof. For a monomial use $\binom{d}{j}\binom{j}{k}/\binom{d}{k}=\binom{d-k}{j-k}$ and the binomial theorem to obtain s^k. Linearity gives the expansion. The Bernstein basis is nonnegative and sums to 1, proving the enclosure. This is the classical identity in the cited Volume 16 passage, not a new positivity theorem. □

The independent sign route here forms coefficients on every closed gap left after removing the open root brackets inside the displayed division intervals. It gives strictly same-sign coefficients on 20, 19, 9, and 6 gaps respectively. No subdivision is needed for this particular data. Their signs, in increasing interval order, are

polynomial	gap signs
A_{X_2}	$-,+,-,+,-,+,-,+,-,+,\quad +,-,+,-,+,-,+,-,+,-$
A_{Y_2}	$+,-,+,-,+,-,-,+,-,-,+,-,-,+,-,+,-,+,-$
A_{X_1}	$+,-,-,+,+,-,+,+,-$
A_{Y_1}	$+,-,+,-,+,-$

On each of the 43 closed root brackets the derivative A' also has strictly same-sign Bernstein coefficients. Combined with the exact endpoint sign change, this proves precisely one simple numerator root per bracket. Denominator nonvanishing then gives precisely the derivative zeros. This route does not use Descartes' rule. The evidence stores every gap's rational endpoints and coefficient extrema, and all four integer coefficient lists; the verifier reconstructs the full coefficients.

8.1 A quadratic allowance at a stationary point

Lemma 8.2 (Stationary-point bracket bound)**.** *Let* $F\in C^2([a,b])$ *with* $|F''|\le M$ *throughout* $[a,b]$ *and* $F'(r)=0$ *for some* $r\in[a,b]$. *Then*

$$F(r)\le F(a)+\tfrac{1}{2}M(b-a)^2. \tag{8.1}$$

Proof. For $a\le t\le r$, $|F'(t)|=|F'(t)-F'(r)|\le M(r-t)$. Integrating from a to r proves $|F(r)-F(a)|\le M(r-a)^2/2\le M(b-a)^2/2$. The root need not be known more accurately than its bracket. □

Proposition 8.3 (Uniform curvature on these 43 brackets)**.** *For both cases and every whole listed bracket,* $|X_\kappa''|<10^9$ *and* $|Y_\kappa''|<10^9$. *Consequently the allowance in* (8.1) *is at most* $2\cdot10^{-11}$ *per stationary point.*

Proof. Every bracket stays at distance greater than $\delta = 7/10000$ from 0 and 1. On $[-1,1]$, $|T_j'| \le j^2$ follows from $T_j' = jU_{j-1}$ and $|U_{j-1}| \le j$. The elementary cosine expansion of U_{k-1} has nonnegative coefficient mass k and degrees at most $k-1$, giving $|U_{k-1}'| \le k(k-1)^2$. Differentiating (7.2), with all coefficient moduli less than one, yields the convenient rational bounds

$$\begin{aligned} |X_\kappa''| &\le \frac{23}{48\delta^2} + 4\sum_{k=1}^{10} k(k-1)^2 + \sum_{k=1}^{10}(k-1) + 3 + 40 \\ &\quad + \frac{80}{(.06)^4} + \frac{13}{(.016)^2} < 10^9, \\ |Y_\kappa''| &\le \frac{11}{48\delta^2} + 2\sum_{k=1}^{10} k(k-1)^2 + \frac{52}{(.016)^4} < 10^9. \end{aligned}$$

The rational $x/(1+x^2)$ derivative has modulus at most one; its coefficient is less than 3. The multiplier derivative has modulus at most $16\lambda_1 < 40$. A T-tail second derivative costs at most $2|r_z||z|^2/(1-|z|)^4$, and an S-tail at most $|r_z||z|^2/(1-|z|)^2$. The tail counts and base bounds give exactly the terms displayed. Case 1 has fewer polynomial terms and no tails. Since $b - a = 2/10^{10}$, half the curvature bound times its square is $1/50000000000$. □

8.2 Values, endpoints, and the improved energy

The exact rational log procedure verifies all 50 upper bounds printed in the source. There are 43 left bracket endpoints and 7 finite-valued division points: -1 for X_2, $1/4, 1/2, 3/4$ for Y_2, and $-1, -1/2, 1/2$ for X_1. None are omitted. The largest relevant point bounds and the norm bounds are

case	quantity	point $10^{10}x$	rational upper bound
2	X	1531948062	$-.984033385315$
2	Y	176402802	-1.608946411646
2	$2\|p\|_*^2 + \|v\|_*^2/2$	—	.778415976284
1	X	2543808026	-1.324655807046
1	Y	532669786	-1.428286151250
1	$\|p\|_*^2 + \|v\|_*^2/2$	—	.931985203901

The exact intervals for every candidate and each norm, not only these maxima, are in `evidence/barrier-check-results.json`. Both X functions tend to $-\infty$ at $0^\pm$ and 1^-; both Y functions tend to $-\infty$ at $0^+, 1^-$. The tangent and multiplier terms are bounded on the closed domains. The finite X endpoint -1 has been included, so exhaustion of stationary points plus the value bounds certifies global suprema.

Theorem 8.4 (Improved same-trial barrier certificate)**.** *For the exact unchanged trial sequences above,*

$$\mathcal{B}_2 < -2.29110250727, \qquad \mathcal{B}_1 < -2.29749544098. \tag{8.2}$$

In particular the maximum barrier upper bound improves on the source's -2.290939875 *by more than* 0.00016263227.

Proof. The rational log calculation gives $\log 2 > .693147180559945$. Use the preceding table and add $2\cdot 10^{-11}$ to each of the two point maxima as allowed by Proposition 8.3. Exact rational addition

yields

$$\begin{aligned}\mathcal{B}_2 &< -\tfrac{11}{16}(.693147180559945) + .778415976284 - .984033385315 - 1.608946411646 + 4\cdot 10^{-11}\\ &< -2.29110250727,\\ \mathcal{B}_1 &< -\tfrac{11}{16}(.693147180559945) + .931985203901 - 1.324655807046 - 1.428286151250 + 4\cdot 10^{-11}\\ &< -2.29749544098.\end{aligned}$$

These are rational strict comparisons. The source instead used a common 120000 first-derivative bound times width $2\cdot 10^{-10}$, costing 0.000024 per supremum, together with coarser norm, point, and log cutoffs. The present result improves that final bounded ledger, not the trial optimization problem itself. □

Corollary 8.5 (Explicitly bounded analytic consequence)**.** *The real-place estimates* (5.6) *and Proposition 6.1, with the same rows and domains, imply*

$$\limsup_{N\to\infty}\left((48N)^{-2}\log|\Delta_N| - \tfrac{1}{2}\log 2\right) \le -2.29110250727.$$

This assertion does not require $G\in\mathbb{Q}$ and does not imply $\Delta_N\neq 0$. The value $\log 0 = -\infty$ is allowed.

9 Stability of the Finite Certificate and Conditional Promotion

Proposition 9.1 (A perturbation budget for trial data)**.** *Fix either certified case, and perturb by real exponentially decaying sequences $\delta p, \delta v$ with $\sum|\delta p_k| \le \epsilon_p$, $\sum|\delta v_k| \le \epsilon_v$. In case 1, also allow $\lambda \mapsto \lambda + \delta\lambda \ge 0$; case 2 must retain $\lambda = 0$. Then*

$$\mathcal{B}_\kappa(p+\delta p, v+\delta v, \lambda+\delta\lambda) \le \mathcal{B}_\kappa(p,v,\lambda) + 12\epsilon_p + 5\epsilon_v + 2\epsilon_p^2 + \tfrac{1}{2}\epsilon_v^2 + \tfrac{5}{6}|\delta\lambda|. \tag{9.1}$$

The last term is zero in case 2.

Proof. For $|x|\le 1$, $|T(\delta u,x)|, |S(\delta u,x)| \le \sum|\delta u_k|$, and $\|\delta u\|_* \le \sum|\delta u_k|$. Both certified weighted norm sums are less than 1, so $\|p\|_* < 2$, $\|v\|_* < 2$ for both cases. Cauchy–Schwarz bounds the norm increase by $\kappa(4\epsilon_p + \epsilon_p^2) + 2\epsilon_v + \epsilon_v^2/2$. The two suprema increase by at most $2\kappa\epsilon_p + 3\epsilon_v + (5/6)|\delta\lambda|$, since $-5/6 \le D \le 1/6$. Use $\kappa \le 2$. No stationary-point persistence assumption is required because this is a uniform function perturbation bound. □

For example $\epsilon_p, \epsilon_v, |\delta\lambda| \le 10^{-6}$ give an increase less than 0.000017834, keeping the larger bound below -2.29108467, still below the source's -2.290939875. This concerns the whole sequence error, not just entrywise rounding of finitely many coefficients while leaving uncontrolled tails.

If a tail of u is discarded after $K \ge d$, then

$$\sum_{k>K}|u_k| \le \sum_z \frac{|r_z||z|^{K+1}}{1-|z|}.$$

Thus truncation has a concrete input to (9.1); it does not permit dropping the slow .984 tail at an arbitrary small order. For the supplied case 2 data, the cruder bounds $10(.94)^{K+1}/.06$ and $13(.984)^{K+1}/.016$ suffice as conservative budgets. No execution-time improvement is claimed for truncation.

Proposition 9.2 (Conditional irrationality bridge)**.** *Assume that under $G\in\mathbb{Q}$ the source's arithmetic estimates establish* (4.3) *along nonzero determinants and establish $\Delta_p \neq 0$ for an unbounded prime sequence. Then Corollary 8.5 contradicts $G\in\mathbb{Q}$.*

Proof. Along that same sequence the lower limit is greater than -2.29084, whereas the upper limit is at most -2.29110250727. The former is strictly larger, contradicting $\liminf \le \limsup$. □

The bound improvement cannot supply either hypothesis in this proposition. If a determinant vanishes, its real estimate is still true and the product formula for nonzero rationals is inapplicable. Likewise, finite contact checks cannot imply a uniform odd-prime valuation statement. The source's hyperbolic-volume deductions, such as irrationality of the two-cusp minimum $4G$ or arithmetic volumes that are positive rational multiples of G, remain dependent on its irrationality conclusion and the cited geometric volume formulas. They are not independent results of the bounded certificate.

10 Current Primary Public Literature

Public-only checks on 8 October 2026 retrieved current repository main as

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`,

with commit date 6 October 2026, 21:58:50 UTC. All nine pinned public TeX files matched the local bytes exactly. The independent review authenticated the two additional ancillary files as well. The exported local directory is not assumed to be a Git checkout. Repository identity is not mathematical acceptance, and no Lean build or formal proof audit is claimed here.

Zudilin's determinantal criterion [2] uses a positive moment representation, controlled denominators, and an explicit size inequality. Its proof uses a squared Vandermonde and positivity for nonvanishing. Its Catalan discussion identifies the denominator obstruction for the approximants considered there. That criterion is a genuine precedent but does not apply automatically to the present signed mixed matrix. The actual proposition and positive-moment proof were inspected.

Lai–Zhou's Theorem 6.1 [3], in the current arXiv v2, states that at least one of

$$\beta(2),\ \beta(4),\ \beta(6),\ \beta(8),\ \beta(10)$$

is irrational. Its original statement and argument were inspected. A finite-family conclusion does not select $\beta(2)$. The independent review also checked the original earlier Rivoal–Zudilin seven-value and Zudilin six-value statements; the larger families must not be confused with a theorem about one individual constant.

Eskandari's September 2026 preprint [4] gives, in Theorem 1.1, an explicit eventual estimate $0 < |G - p_n/q_n| \le q_n^{-0.62}$. The statement and its explanation of the denominator exponent were inspected in the primary text. Multiplying by q_n gives only an upper bound $q_n^{0.38}$, not a sequence of nonzero integer forms tending to zero. This is neither an irrationality measure nor an irrationality theorem. The author metadata for Eskandari–Murty–Nemoto [5] was also checked for context; no theorem from that paper is imported here.

Calegari–Dimitrov–Tang's conductor-3 independence theorem [6] concerns $1, \zeta(2), L(2, \chi_{-3})$, not $G = L(2, \chi_{-4})$. The independent review checked its primary Theorem A but did not audit the entire proof. Similarly Calegari's 2-adic result concerns a value in $\mathbb{Q}_2$, not an identification of the real constant with that value. The present smoothing argument introduces two discrepancy constants and does not assume such an identification.

Sun's September 2026 manuscript [7] claims the same qualitative irrationality conclusion. Wachs's primary critique [8] concerns Sun's bounding method and identifies a missing leading 2-adic term in its ledger. The critique does not conclude that G is rational, and it is not a refutation of the different pinned OpenAI construction. The primary statement and critique were checked as current context; Wachs's computations were not replayed for this report. No result from Sun is used in the local proofs above.

For the methods actually used, Chebyshev conventions agree with NIST DLMF 18.5 [9]; determinant integration and alternants are classical [10, 11]; the Chebyshev logarithmic-kernel identity is proved here directly from a power series. Primary literature status checks and finite certificates are kept distinct from global proof review. No confidential title, passage, or private hash was included in a network request.

11 Benefits, Limitations, and Future Mathematics

11.1 What has actually improved

The completed contribution is a source-specific, exact finite certificate and a tighter bounded analytic constant with the original trial sequences unchanged. The contact mechanism is explained as a finite moment map on two explicitly different polynomial spaces. The Bernstein route gives independent root exhaustion, including the sign of cleared denominators; the curvature lemma charges the entire bracket, not merely sampled values. The perturbation proposition supplies an explicit stability budget and requires tail error to be included.

The new bound is not asserted to be the optimal value of the trial functional, and the given sequences were not discovered or reoptimized here. The improvement comes from retaining already available precise point and norm values and sharpening the interpolation allowance. Bernstein enclosures, Descartes' rule, Taylor estimates, and square completion are classical methods. Their source-specific assembly and replay are the contribution, not new general mathematics in those tools.

11.2 Pros and cons of the Ram viewpoint

The principal benefit is explicit bookkeeping: it identifies the physical finite spaces, the functionals actually measured, the coefficient contact that eliminates $\zeta(2)$, and the exact polynomial sign problem left after all denominator signs have been fixed. All numerical decisions in the final barrier certificate are rational inequalities with reproducible data. The representation makes it easier to see which statements depend on a rank assumption and which hold without one.

The principal limitation is that changing coordinates does not establish rank, positivity, all-prime integrality, or an asymptotic comparison. The determinant is signed; local jets do not create a positive Gram matrix. Endpoint singularities require uniform damping, diagonal bounds, and a justified order of limits. A finite bank cannot provide a uniform inverse on an unbounded sequence of growing spaces without additional estimates. Slow geometric tails can also make naive polynomial truncation costly. Exact rational verification may be slower than floating-point exploration; no speedup is claimed.

11.3 A focused next program

First, optimize the dual trials with a posteriori rational certification, retaining both case domains and the multiplier sign constraint. A better candidate is not a better certified bound until all roots and endpoints are controlled. Second, adaptively use local curvature or rational Bernstein bounds on A/Q to reduce certificate size; compare actual runtime and bit complexity before claiming efficiency.

Third, quantify tail truncation via (9.1), possibly finding a simpler fully polynomial trial with the same needed margin. Fourth, seek explicit finite-N constants in the energy regularization and arithmetic errors. The present limsup argument does not give an effective threshold or irrationality measure. Finally, a full proof-validation program must audit every arithmetic-to-asymptotic and real-place bridge, including any formalized statements and their dependency assumptions. An independently replayed finite certificate is one part of that program, not its completion.

Contribution score. Applicability is L4 because the finite-moment and polynomial-enclosure interfaces are substantial and concrete. Contribution is C2/5: Theorems 8.4 and Proposition 9.1 establish bounded certificate refinements and reproducibility with clear limits. There is no claim of a new global irrationality proof, unconditional acceptance of the source theorem, new optimal trial coefficients, an irrationality measure, or computational speedup. A C3 assessment is not forced by the importance of the source's title.

A Read Scope, Tests, and Bounded Reviews

A.1 Exact reads and confidential interfaces

The nine mathematical source files have respective line counts 44 for `main.tex`, 208 for introduction, 434 for foundations, 672 for odd primes, 413 for nonvanishing, 420 for real place, 288 for energy, 471 for certificate, and 51 for conclusion. All were read in full. The 17-line README and 319-line bibliography were also read. Read depth and executable coverage are distinct: the report's direct certificate replay covers the analytic barriers and finite contact identities; the separate independent review covers the arithmetic chain in detail.

The actual private locations, recorded with full local hashes and ranges in `evidence/read-provenance.json`, are:

- `RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`, lines 140–226 and 442–551, including the actual finite representation and polynomial positivity passages.
- `RamCalBooks/volumes/volume16/chapters/ch05d-clearance.tex`, lines 1–143, including `v16c05:conversionproof`, `v16c05:hull`, and exact subdivision.
- `RamCalBooks/volumes/volume1-rev2/chapters/ch03-local-moments-rev2.tex`, lines 1–158, including `thm:ram-local`, tensor terms, and the source-centered product expansion.

These are local, unpublished, author-supplied references. Naming a passage is not a substitute for reading it; the actual statements and proofs were used in the corresponding interfaces above.

A.2 Exact author tests and their mathematical scope

The reproducible script `evidence/author_checks.py` uses Python 3.10 and SymPy for exact polynomial algebra, with standard-library rational interval arithmetic for logs. It checks the following finite statements:

- Degrees $36, 24, 13, 9$; all Descartes variation counts; all 43 exact bracket sign changes; 54 strictly signed Bernstein gap enclosures; strict numerator-derivative sign on every bracket.
- All 50 printed point upper bounds and both printed norm upper bounds, with explicit rational logarithm and argument remainders.
- Tail moduli and coefficient inequalities; the curvature bound; the two improved rational energy comparisons; the perturbation example; the negative-square completion identities.
- Every row support and every contact coefficient $0 \le j < L$ at $N = 1, 2, 3$: 288 rows and 39,648 exact equalities. These corroborate the all-N algebraic proof, rather than replace it.

The output `evidence/barrier-check-results.json` contains exact rational interval endpoints, primitive numerator coefficients, gap certificates, all candidate values, and the final test status. There is no numerical root finder and no floating-point logarithm in the certification path. Floating-point fields in the JSON are labeled diagnostic displays only. A depth-exhausted Bernstein test raises an inconclusive error rather than reporting success.

A.3 Independent arithmetic review

The completed independent review found no concrete blocking error within its focused foundations, finite-prime arithmetic, and conditional nonvanishing scope. It explicitly did not certify the global theorem. It read all eleven source and ancillary files, authenticated their bytes at the public pin, and distinguished its detailed arithmetic audit from its dependency reading of the energy and barrier sections.

Its separate exact replay reports 212,334 M and 212,334 Z digit congruences at eight permitted (p, H) pairs, including 12,285 negative boundary-index cases; 41,405 M and 41,405 Z central-layer tests; all 144 published pivots; fixed determinants modulo 101, 103, and 107; palindromic bases and Frobenius polynomial identities at $p = 3, 5, 7, 11$; 82,875 algebraic counting/loss evaluations; exact loss integration and the 505/4608 coefficient. Small-prime polynomial identities were not presented as tests of the $p > 260$ moment bridge.

Its 2-adic replay checks 81 smoothing/discrepancy congruences modulo 2^{128} at indices in $\{0, 1, 2, 18, 19, 32, 48, 58, 64\}$. The placeholder 7/13 tests the rational parameter identity only and is not assigned as a value of G. Terms $k \geq 300$ have a proved tail bound of at least 283 bits, or 219 bits after the worst kernel loss; the observed discrepancy valuation is at least 234. These are bounded checks, not empirical proofs of an all-index convergence or determinant asymptotic.

The review also inspected the $p \nmid \mathrm{den}(G)$ qualification, the integral column shears, the distinct extraction shifts, the auxiliary row of lower degree, and the fixed exceptional prime set in the nonvanishing bridge. It did not replay the barrier certificate, audit all cited literature proofs, verify Lean, or validate the hyperbolic volume corollaries. Those exclusions remain part of its bounded verdict. The report incorporates its actual results with attribution, not as additional author executions.

Completed full-manuscript review. A full mathematical read of this report was completed after the independent arithmetic review. It checked the moment/contact and finite-space arguments, the uniform energy endpoint and damping route, the exact trial tables against source data, and the entire certificate implementation. Separate rational calculations confirmed curvature bounds below 10^9 and the two displayed improved energy comparisons. The literature, limitations, reproduction, and reference sections were also read. No decisive mathematical defect was found in that bounded review; this is not global certification of the source irrationality theorem.

A.4 Reproduction and immutable evidence

The mathematical replay command is

```
py -3.10 evidence/author_checks.py
```

from the report directory. Public pin authentication is reproduced by `evidence/collect_provenance.py`, whose network requests are restricted to fixed public repository URLs. Build diagnostics, source hashes, read ranges, exact test outputs, and bounded review receipts accompany the report. The frozen manifest covers the final TeX, PDF, and copied immutable evidence, excluding shared mutable files and the manifest's own hash. These records establish artifact identity and bounded checks, not a theorem about global source correctness.

Physical build. The native editor-open request returned a queued result. The native compiler failed with `Unable to find standard directories for platform`. The existing physical exporter separately compiled the standalone source without undefined references. A first 22-page export identified two overlong prose lines; the final format pass wraps those lines and separates the contents from the title page. Final compiler logs, page count, and all-page visual inspection evidence are stored with the report.

References

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[2] W. Zudilin, *A determinantal approach to irrationality*, Constructive Approximation 45 (2017), 301–310; Proposition 2 and Section 6. https://arxiv.org/html/1507.05697v2.

[3] L. Lai and L. Zhou, *At least two of* $\zeta(5), \zeta(7), \ldots, \zeta(35)$ *are irrational*, Publicationes Mathematicae Debrecen 101 (2022), 353–372; Theorem 6.1. https://arxiv.org/html/2103.00904v2.

[4] P. Eskandari, *Improved rational approximations to Catalan's constant*, 22 September 2026, Theorem 1.1. https://arxiv.org/html/2609.26354v1.

[5] P. Eskandari, V. K. Murty and Y. Nemoto, *Mixed motives and linear forms in the Catalan constant*. https://arxiv.org/abs/2510.20648. Metadata checked for context only.

[6] F. Calegari, V. Dimitrov and Y. Tang, *The linear independence of* 1, $\zeta(2)$, *and* $L(2, \chi_{-3})$, Theorem A. https://arxiv.org/abs/2408.15403.

[7] Z.-W. Sun, *Catalan's constant is irrational*, 3 September 2026, preprint claim. https://arxiv.org/abs/2609.04176v1.

[8] D. Wachs, *A note on a recent claimed proof of the irrationality of Catalan's constant*, 16 September 2026, critique of Sun's different construction. https://arxiv.org/html/2609.22339v1.

[9] NIST, *Digital Library of Mathematical Functions*, Section 18.5, equations 18.5.1–18.5.2. https://dlmf.nist.gov/18.5.

[10] P. J. Forrester, *Meet Andréief, Bordeaux 1886, and Andreev, Kharkov 1882–83*. https://arxiv.org/abs/1806.10411v1.

[11] C. Krattenthaler, *Advanced determinant calculus*, Séminaire Lotharingien de Combinatoire 42 (1999), B42q. https://arxiv.org/abs/math/9902004v3.

[12] M. Subbarao, revised TR250 and the Volume 16 and Volume 1 passages identified in Appendix A; confidential author-supplied mathematical corpus, used locally only.